INITIAL FITTING IDEALS OF FINITE-LAYER ANTICYCLOTOMIC SELMER GROUPS VIA THETA ELEMENTS

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

1. Introduction

1.1. **The statement of the main result.** Let E be an elliptic curve of conductor N over $\mathbb{Q}$ and $p \geq 5$ be a prime of good ordinary reduction for E such that

(Im) the mod p Galois representation $\overline{\rho} : G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{Aut}_{\mathbb{F}_p}(E[p]) \simeq \mathrm{GL}_2(\mathbb{F}_p)$ is surjective, and

(Ram) $\overline{\rho}$ is ramified at every prime dividing N, so p does not divide Tamagawa factors of E.

Let K be an imaginary quadratic field of odd discriminant $-D_K < -4$ with $(D_K, Np) = 1$ such that

(Spl) p splits in K, and

(Na) $a_p(E) \not\equiv 1 \pmod{p}$.

Write

$$N = N^+ \cdot N^-$$

where a prime divisor of N^+ splits in K and a prime divisor of N^- is inert in K.

(Def) Assume that N^- is a square-free product of an odd number of primes.

Let K_∞ be the anticyclotomic $\mathbb{Z}_p$-extension of K and K_n be the subextension of K in K_∞ of degree p^n for $n \geq 0$. Let $\Lambda_n = \mathbb{Z}_p[\mathrm{Gal}(K_n/K)] \simeq \mathbb{Z}_p[X]/((1+X)^{p^n} - 1)$ be the finite layer Iwasawa algebra. Under (Def), denote by

$$\theta(E/K_n) = \sum_{\sigma \in \mathrm{Gal}(K_n/K)} a_\sigma \cdot \sigma \in \Lambda_n$$

Bertolini–Darmon's theta element of E over K_n which interpolates the square-roots of $L(E/K, \chi, 1)$ for finite order characters χ on $\mathrm{Gal}(K_n/K)$. It is reviewed in §2. For the natural projection map $\pi_{n,n-1} : \Lambda_n \to \Lambda_{n-1}$,

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let $\nu_{n-1,n} : \Lambda_{n-1} \to \Lambda_n$ be the map defined by $\sigma \mapsto \sum_{\pi_{n,n-1}(\tau)=\sigma} \tau$. For $0 \leq m \leq n$, we write $\nu_{m,n} = \nu_{n-1,n} \circ \nu_{n-2,n-1} \circ \cdots \circ \nu_{m,m+1}$. Then we have the equality of ideals of Λ_n (Lemma 2.1)

$$(\theta(E/K_n), \nu_{n-1,n}(\theta(E/K_{n-1}))) = (\nu_{m,n}(\theta(E/K_m)) : 0 \leq m \leq n),$$

and it is a principal ideal under (Na) (Lemma 2.2).

The goal of this article is to prove the following anticyclotomic analogue of the "strong" main conjecture of Kurihara [Kur02, Conj. 0.3], which refines the "weak" main conjecture of Mazur and Tate [MT87, Conj. 3].

Theorem 1.1. *Under the assumptions mentioned above, i.e. (Im),(Ram),(Spl),(Na), and (Def), the theta elements over K_m with $0 \leq m \leq n$ generate the initial Fitting ideal of dual Selmer groups over K_n, i.e. we have equality of ideals of Λ_n*

$$(\theta(E/K_n), \nu_{n-1,n}(\theta(E/K_{n-1})))^2 = \mathrm{Fitt}_{\Lambda_n}\left(\mathrm{Sel}(K_n, E[p^\infty])^\vee\right),$$

which is indeed a principal ideal, where $\mathrm{Sel}(K_n, E[p^\infty])$ is the classical Selmer group of $E[p^\infty]$ over K_n and $(-)^\vee$ means the Pontryagin dual.

Since theta elements interpolate the *square-roots* of twisted Rankin–Selberg L-values, it is natural that the *square* of the ideal generated by theta elements appears in the equality.

The strategy of our proof follows that given in [KK21] and we also add some details on the "p-destabilization" process and on the comparison of various anticyclotomic Selmer groups of elliptic curves.

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2. Bertolini–Darmon's theta elements and anticyclotomic p-adic L-functions

We quickly review the construction of Gross points of conductor p^n, theta elements, and anticyclotomic p-adic L-functions. See [CH15, CH18, Kim19] for details.

2.1. Gross points.

Let K be the imaginary quadratic field of odd discriminant $-D_K < -4$. Define

$$\vartheta := \frac{D_K - \sqrt{-D_K}}{2}$$

so that $\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\vartheta$. Let B_{N^-} be the definite quaternion algebra over $\mathbb{Q}$ of discriminant N^-. Then there exists an embedding $\Psi : K \hookrightarrow B_{N^-}$ [Vig80]. More explicitly, we choose a K-basis $(1, J)$ of B_{N^-} so that $B_{N^-} = K \oplus K \cdot J$ such that $\beta := J^2 \in \mathbb{Q}^\times$ with $\beta < 0$, $J \cdot t = \bar{t} \cdot J$ for all $t \in K$, $\beta \in \left(\mathbb{Z}_q^\times\right)^2$ for all $q \mid pN^+$, and $\beta \in \mathbb{Z}_q^\times$ for all $q \mid D_K$. Fix a square root $\sqrt{\beta} \in \overline{\mathbb{Q}}$ of β. For a $\mathbb{Z}$-module A, write $\widehat{A} = A \otimes \widehat{\mathbb{Z}}$. Fix an isomorphism

$$i := \prod i_q : \widehat{B}_{N^-}^{(N^-)} \simeq \mathrm{M}_2(\mathbb{A}^{(N^-\infty)})$$

as follows:

- For each finite place $q \mid N^+p$, the isomorphism $i_q : B_{N^-,q} \simeq \mathrm{M}_2(\mathbb{Q}_q)$ is defined by

$$i_q(\vartheta) = \begin{pmatrix} \mathrm{trd}(\vartheta) & -\mathrm{nrd}(\vartheta) \\ 1 & 0 \end{pmatrix}, \qquad i_q(J) = \sqrt{\beta} \cdot \begin{pmatrix} -1 & \mathrm{trd}(\vartheta) \\ 0 & 1 \end{pmatrix}$$

  where trd and nrd are the reduced trace and the reduced norm on B, respectively.
- For each finite place $q \nmid pN^+$, the isomorphism $i_q : B_{N^-,q} \simeq \mathrm{M}_2(\mathbb{Q}_q)$ is chosen so that $i_q\left(\mathcal{O}_K \otimes \mathbb{Z}_q\right) \subseteq \mathrm{M}_2(\mathbb{Z}_q)$.

Under the fixed isomorphism i, for any rational prime q, the local Gross point $\varsigma_q \in B_{N^-,q}^\times$ is defined as follows:

- $\varsigma_q := 1$ in $B_{N^-,q}^\times$ for $q \nmid pN^+$.
- $\varsigma_q := \frac{1}{\sqrt{D_K}} \cdot \begin{pmatrix} \vartheta & \overline{\vartheta} \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(K_\mathfrak{q}) = \mathrm{GL}_2(\mathbb{Q}_q)$ for $q \mid N^+$ with $q = \mathfrak{q}\overline{\mathfrak{q}}$ in $\mathcal{O}_K$.
- $\varsigma_p^{(n)} = \begin{pmatrix} \vartheta & -1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} p^n & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(K_\mathfrak{p}) = \mathrm{GL}_2(\mathbb{Q}_p)$ where $p = \mathfrak{p}\overline{\mathfrak{p}}$ splits in K.

Let $\widehat{\Psi} : \widehat{K} \hookrightarrow \widehat{B}_{N^-}$ be the adelic version of Ψ. We define $x_n : \widehat{K}^\times \to \widehat{B}_{N^-}^\times$ by $x_n(a) = \widehat{\Psi}(a) \cdot \varsigma^{(n)} := \widehat{\Psi}(a) \cdot \left(\varsigma_p^{(n)} \times \prod_{q \neq p} \varsigma_q\right)$. The collection $\left\{x_n(a) : a \in \widehat{K}^\times\right\}$ of points is called the **Gross points of conductor p^n on** $\widehat{B}_{N^-}^\times$. The fixed embedding $K \hookrightarrow B_{N^-}$ also induces an optimal embedding of $\mathcal{O}_n = \mathbb{Z} + p^n\mathcal{O}_K$ into the Eichler order $B_{N^-} \cap \varsigma^{(n)} \widehat{R}_{N^+} (\varsigma^{(n)})^{-1}$ where R_{N^+} is the Eichler order of level N^+ under the fixed isomorphism i.

2.2. **Theta elements.** Let $f(z) = \sum_{n \geq 1} a_n q^n \in S_2(\Gamma_0(N))$ be the cuspidal newform of weight two with rational Fourier coefficients corresponding to E via the modularity theorem [BCDT01]. Let $\phi_f : B_{N^-}^\times \backslash \widehat{B}_{N^-}^\times / \widehat{R}_{N^+}^\times \to \mathbb{C}$ be the Jacquet–Langlands transfer of f. Since $B_{N^-}^\times \backslash \widehat{B}_{N^-}^\times / \widehat{R}_{N^+}^\times$ is a finite set and f is a Hecke eigenform, we are able to and do normalize

$$\phi_f : B_{N^-}^\times \backslash \widehat{B}_{N^-}^\times / \widehat{R}_{N^+}^\times \to \mathbb{Z}_p$$

such that the image of ϕ_f does not lie in $p\mathbb{Z}_p$. This integral normalization is related to the congruence ideals [PW11, Kim17, KO23]. Let

$$\widetilde{\theta}_n(E/K) = \sum_{[a]\in\mathcal{G}_n} \phi_f(x_n(a)) \cdot [a] \in \mathbb{Z}_p[\mathcal{G}_n]$$

where $\mathcal{G}_n = K^\times \backslash \widehat{K}^\times / \widehat{\mathcal{O}}_n^\times$ and $[a]$ is the image of $a \in \widehat{K}^\times$ in $\mathcal{G}_n$. Then **Bertolini–Darmon's theta element** $\theta(E/K_n)$ **of** E **over** K_n is defined by the image of $\widetilde{\theta}_n(E/K)$ in Λ_n

$$\begin{aligned} \mathbb{Z}_p[\mathcal{G}_n] &\longrightarrow \Lambda_n = \mathbb{Z}_p[\mathrm{Gal}(K_n/K)] \\ \widetilde{\theta}_n(E/K) &\longmapsto \theta(E/K_n) \end{aligned}$$

where the map is naturally induced from the quotient map $\mathcal{G}_n \to \mathrm{Gal}(K_n/K)$. It is known that $\theta(E/K_n)$ interpolates "an half of" $L(E,\chi,1)$ where χ runs over characters on $\mathrm{Gal}(K_n/K)$. See [CH18, Rem. (iii) after Thm. A] for the precise meaning of "an half of". Because $\theta(E/K_n)$ depends on the choice of Gross points, $\theta(E/K_n)$ is well-defined only up to multiplication by $\mathrm{Gal}(K_n/K)$.

2.3. ***p*-adic *L*-functions.** Let α, β be the roots of the Hecke polynomial $X^2 - a_p X + p$ of f at p. Since f is ordinary at p, one of them, say α, is a p-adic unit.

The p**-stabilization** $f_\alpha \in S_2(\Gamma_0(Np))$ **of** f is defined by

$$f_\alpha(z) = f(z) - \beta \cdot f(pz)$$

whose U_p-eigenvalue is α. Then the **theta element of** f_α **over** K_n is characterized by the following relation:

$$\theta(f_\alpha/K_n) = \frac{1}{\alpha^n} \cdot \left(\theta(E/K_n) - \frac{1}{\alpha} \cdot \nu_{n-1,n}(\theta(E/K_{n-1})) \right). \tag{2.1}$$

It is known that theta elements of E satisfies the three term relation (e.g. [DI08, Lem. 2.6])

$$\pi_{n+1,n}\left(\theta(E/K_{n+1})\right) = a_p \cdot \theta(E/K_n) - \nu_{n-1,n}\left(\theta(E/K_{n-1})\right) \tag{2.2}$$

and the theta elements of f_α satisfy the norm compatibility

$$\pi_{n+1,n}\left(\theta(f_\alpha/K_{n+1})\right) = \theta(f_\alpha/K_n). \tag{2.3}$$

Let ι be the involution on Λ_n defined by inverting group-like elements, so we have

$$\iota\Big(\sum_{\sigma\in\mathrm{Gal}(K_n/K)} a_\sigma \cdot \sigma\Big) = \sum_{\sigma\in\mathrm{Gal}(K_n/K)} a_\sigma \cdot \sigma^{-1}.$$

We define the **anticyclotomic** p**-adic** L**-function of** E by

$$L_p(E/K_\infty) = \varprojlim_n \left(\theta(f_\alpha/K_n) \cdot \iota(\theta(f_\alpha/K_n))\right) \in \Lambda = \varprojlim_n \Lambda_n.$$

This element is well-defined. The functional equation for Bertolini–Darmon's theta elements yields the equality of ideals of Λ (e.g. [BD96, Prop. 2.13], [BD05, Lem. 1.5])

$$(\theta(f_\alpha/K_n)) = (\iota(\theta(f_\alpha/K_n)))\,. \tag{2.4}$$

We prove two useful lemmas.

Lemma 2.1. *We have an equality of ideals of* Λ_n

$$(\theta(E/K_n), \nu_{n-1,n}(\theta(E/K_{n-1}))) = (\nu_{m,n}(\theta(E/K_m)) : 0 \le m \le n)\,.$$

Proof. From the three term relation (2.2), we have

$$\nu_{n-1,n}(\pi_{n,n-1}(\theta(E/K_n))) = a_p \cdot \nu_{n-1,n}(\theta(E/K_{n-1})) - \nu_{n-2,n}(\theta(E/K_{n-2}))$$

for $n \ge 2$. Since $\nu_{n-1,n}(\pi_{n,n-1}(\theta(E/K_n))) = f_n \cdot \theta(E/K_n)$ for some $f_n \in \Lambda_n$, we have

$$\nu_{n-2,n}(\theta(E/K_{n-2})) \in (\theta(E/K_n), \nu_{n-1,n}(\theta(E/K_{n-1}))) \subseteq \Lambda_n.$$

In the same manner, we can obtain

$$\nu_{n-3,n-1}(\theta(E/K_{n-3})) \in (\theta(E/K_{n-1}), \nu_{n-2,n-1}(\theta(E/K_{n-2}))) \subseteq \Lambda_{n-1}.$$

By taking $\nu_{n-1,n}$, we have

$$\begin{aligned}\nu_{n-3,n}(\theta(E/K_{n-3})) &\in (\nu_{n-1,n}(\theta(E/K_{n-1})), \nu_{n-2,n}(\theta(E/K_{n-2})))\\ &\subseteq (\theta(E/K_n), \nu_{n-1,n}(\theta(E/K_{n-1})))\\ &\subseteq \Lambda_n.\end{aligned}$$

By applying this argument recursively, the conclusion follows. □

Lemma 2.2. *Under (Spl) and (Na), we have an equality of ideals of* Λ_n

$$(\theta(E/K_n), \nu_{n-1,n}(\theta(E/K_{n-1}))) = (\theta(f_\alpha/K_n))\,.$$

Proof. By the definition of the p-stabilization (2.1), we have one inclusion $\supseteq$. Hence, we focus on the opposite inclusion. By the interpolation formula of the anticyclotomic p-adic L-functions [CH18, Thm. A] under (Spl), we have the comparison of (the square-roots of) L-values

$$\theta(f_\alpha/K) = \left(1 - \frac{1}{\alpha}\right) \cdot \theta(E/K).$$

Under (Na), we have equality in $\mathbb{Z}_p$

$$\left(1 - \frac{1}{\alpha}\right)^{-1} \cdot \theta(f_\alpha/K) = \theta(E/K),$$

so we have $(\theta(E/K)) \subseteq (\theta(f_\alpha/K))$. In fact, they are the same ideal. From (2.1) and (2.3), we have

$$\begin{aligned}
\theta(f_\alpha/K_1) &= \frac{1}{\alpha} \cdot \left(\theta(E/K_1) - \frac{1}{\alpha} \cdot \nu_{0,1}(\theta(E/K)) \right) \\
&= \frac{1}{\alpha} \cdot \left(\theta(E/K_1) - \frac{1}{\alpha} \cdot \nu_{0,1}(\left(1 - \frac{1}{\alpha}\right)^{-1} \cdot \theta(f_\alpha/K)) \right) \\
&= \frac{1}{\alpha} \cdot \left(\theta(E/K_1) - \frac{1}{\alpha} \cdot \left(1 - \frac{1}{\alpha}\right)^{-1} \cdot \nu_{0,1}(\pi_{1,0}(\theta(f_\alpha/K_1))) \right) \\
&= \frac{1}{\alpha} \cdot \left(\theta(E/K_1) - \frac{1}{\alpha} \cdot \left(1 - \frac{1}{\alpha}\right)^{-1} \cdot f_1 \cdot \theta(f_\alpha/K_1) \right)
\end{aligned}$$

for some $f_1 \in \Lambda_1$. This shows that $\theta(E/K_1) = g_1 \cdot \theta(f_\alpha/K_1)$ for some $g_1 \in \Lambda_1$.

We suppose that $\theta(E/K_{n-1}) = g_{n-1} \cdot \theta(f_\alpha/K_{n-1})$ for some $g_{n-1} \in \Lambda_{n-1}$.

$$\begin{aligned}
\theta(f_\alpha/K_n) &= \frac{1}{\alpha^n} \cdot \left(\theta(E/K_n) - \frac{1}{\alpha} \cdot \nu_{n-1,n}(\theta(E/K_{n-1})) \right) \\
&= \frac{1}{\alpha^n} \cdot \left(\theta(E/K_n) - \frac{1}{\alpha} \cdot \nu_{n-1,n}(g_{n-1} \cdot \theta(f_\alpha/K_{n-1})) \right) \\
&= \frac{1}{\alpha^n} \cdot \left(\theta(E/K_n) - \frac{1}{\alpha} \cdot g_{n-1} \cdot \nu_{n-1,n}(\pi_{n,n-1}(\theta(f_\alpha/K_n))) \right) \\
&= \frac{1}{\alpha^n} \cdot \left(\theta(E/K_1) - \frac{1}{\alpha} \cdot g_{n-1} \cdot f_n \cdot \theta(f_\alpha/K_n) \right)
\end{aligned}$$

for some $f_n \in \Lambda_n$. This shows that $\theta(E/K_n) = g_n \cdot \theta(f_\alpha/K_n)$ for some $g_n \in \Lambda_n$. By induction, we have inclusion

$$(\theta(E/K_n)) \subseteq (\theta(f_\alpha/K_n)),$$

so we also have

$$(\nu_{n-1,n}(\theta(E/K_{n-1}))) \subseteq (\nu_{n-1,n}(\theta(f_\alpha/K_{n-1}))).$$

Since $\nu_{n-1,n}(\theta(f_\alpha/K_{n-1})) = f_n \cdot \theta(f_\alpha/K_n)$, we have

$$(\nu_{n-1,n}(\theta(f_\alpha/K_{n-1}))) \subseteq (\theta(f_\alpha/K_n)).$$

The conclusion follows. □

3. Comparison of Selmer groups

3.1. **Local properties of Galois representations.** Let $\rho : G_{\mathbb{Q}} \to \mathrm{Aut}_{\mathbb{Q}_p}(V) = \mathrm{GL}_2(\mathbb{Q}_p)$ be the two-dimensional Galois representation associated to E.

- Since E is good ordinary at p, we have

$$\rho|_{G_{\mathbb{Q}_p}} \sim \begin{pmatrix} \chi_\alpha^{-1} \cdot \chi_{\mathrm{cyc}} & * \\ 0 & \chi_\alpha \end{pmatrix}$$

where χ_α is the unramified character sending the arithmetic Frobenius at p to α.

- For ℓ dividing N exactly, we also have

$$\rho|_{G_{\mathbb{Q}_\ell}} \sim \begin{pmatrix} \pm\chi_{\mathrm{cyc}} & * \\ 0 & \pm\mathbf{1} \end{pmatrix}.$$

For a rational prime v dividing N^-p, we consider the following subspaces:

- For $v = p$, let $F^+V \subseteq V$ be the subspace on which the inertia subgroup I_v acts by χ_{cyc}.
- For a rational prime v dividing N^-, $F^+V \subseteq V$ be the subspace on which the inertia subgroup I_v acts by χ_{cyc} or $\chi_{\mathrm{cyc}}\tau_v$ where τ_v is the non-trivial unramified quadratic character of $G_{\mathbb{Q}_v}$.

Let L be an algebraic extension of K. For a prime w of L dividing N^-p, we define the **ordinary local condition of V at w** by

$$\mathrm{H}^1_{\mathrm{ord}}(L_w, V) = \ker\left(\mathrm{H}^1(L_w, V) \to \mathrm{H}^1(L_w, V/F^+V)\right).$$

Denote by $T = \varprojlim_k E[p^k]$ the p-adic Tate module of E, so we have $T \otimes_{\mathbb{Z}_p} \mathbb{Q}_p = V$, and by $E[p^\infty] = \varinjlim_k E[p^k]$ the p-power torsion points of E. Then the same local conditions for T, T/p^kT, $E[p^\infty]$, and $E[p^k]$ are defined by propagation.

3.2. **N^--ordinary (residual) Selmer groups.** Let Σ be the finite set of places of $\mathbb{Q}$ consisting of the places dividing $Np\infty$, and K_Σ be the maximal extension of K unramified outside Σ. We write

- $\Sigma^+ \subseteq \Sigma$ to be the subset of Σ consisting of the places not dividing $p\infty$ which split in $K/\mathbb{Q}$, and
- $\Sigma^- \subseteq \Sigma$ to be the subset of Σ consisting of the places not dividing $p\infty$ which are inert in $K/\mathbb{Q}$.

For a place w of K_∞, we write $w \in \Sigma^\pm$ if w divides a rational prime ℓ contained in $\Sigma^\pm$, respectively. For every $k \geq 1$, we define the **N^--ordinary (and N^+-strict) Selmer group of $E[p^k]$** $\mathrm{Sel}_{N^-}(K_\infty, E[p^k])$ by the kernel of the map

$$\mathrm{H}^1(K_\Sigma/K_\infty, E[p^k]) \to \prod_{w\nmid\Sigma^+} \mathrm{H}^1(K_{\infty,w}, E[p^k]) \times \prod_{w\in\Sigma^- \text{ or } w|p} \frac{\mathrm{H}^1(K_{\infty,w}, E[p^k])}{\mathrm{H}^1_{\mathrm{ord}}(K_{\infty,w}, E[p^k])}$$

and define $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty]) = \varinjlim_k \mathrm{Sel}_{N^-}(K_\infty, E[p^k])$. This is *the* Selmer group used in the bipartite Euler system argument [BD05, Def. 2.8].

3.3. **Minimal and Greenberg Selmer groups.** We follow the convention of [PW11, §3.1]. The **minimal Selmer group** $\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])$ **of** $E[p^\infty]$ is defined by the kernel of the map

$$\mathrm{H}^1(K_\infty, E[p^\infty]) \to \prod_{w\nmid p} \mathrm{H}^1(K_{\infty,w}, E[p^\infty]) \times \prod_{w|p} \frac{\mathrm{H}^1(K_{\infty,w}, E[p^\infty])}{\mathrm{H}^1_{\mathrm{ord}}(K_{\infty,w}, E[p^\infty])},$$

and the **Greenberg Selmer group** $\mathrm{Sel}_{\mathrm{Gr}}(K_\infty, E[p^\infty])$ **of** $E[p^\infty]$ is defined by the kernel of the map

$$\mathrm{H}^1(K_\infty, E[p^\infty]) \to \prod_{w\nmid p} \mathrm{H}^1(I_{\infty,w}, E[p^\infty]) \times \prod_{w|p} \frac{\mathrm{H}^1(K_{\infty,w}, E[p^\infty])}{\mathrm{H}^1_{\mathrm{ord}}(K_{\infty,w}, E[p^\infty])}$$

where $I_{\infty,w}$ is the inertia subgroup of $G_{K_{\infty,w}}$. Under (Ram), $\overline{\rho}$ is ramified at every prime dividing N, so p does not divide any Tamagawa factors. Then by using [PW11, Lem. 3.4], we have an isomorphism

$$\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty]) \simeq \mathrm{Sel}_{\mathrm{Gr}}(K_\infty, E[p^\infty]). \tag{3.1}$$

3.4. **The comparison.** We recall the final displayed equation in the proof of [PW11, Prop. 3.6]:

$$0 \longrightarrow \mathrm{Sel}_{N^-}(K_\infty, E[p^k]) \longrightarrow \mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])[p^k] \longrightarrow \prod_w \frac{(E[p^\infty])^{G_{K_{\infty,w}}}}{p^k (E[p^\infty])^{G_{K_{\infty,w}}}}$$

where w runs over the primes of K_∞ dividing N^+. The local conditions at primes dividing N^- of minimal Selmer groups and N^--ordinary Selmer groups coincide since such primes split completely in K_∞/K. Thus, we have inclusion

$$\mathrm{Sel}_{N^-}(K_\infty, E[p^k]) \subseteq \mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])[p^k]$$

which is of finite index and is independent of k.

Proposition 3.1. *If* $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])$ *is* Λ*-cotorsion with vanishing of* μ*-invariant, then* $\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])$ *is also* Λ*-cotorsion with vanishing of* μ*-invariant.*

Proof. We have $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])[p] = \mathrm{Sel}_{N^-}(K_\infty, E[p])$ since N^--ordinary Selmer groups of $E[p^\infty]$ are defined as the injective limit of N^--ordinary Selmer groups of $E[p^k]$. By the assumption, $\mathrm{Sel}_{N^-}(K_\infty, E[p])$ is finite as noted in the proof of [KPW17, Cor. 2.3]. Since the inclusion $\mathrm{Sel}_{N^-}(K_\infty, E[p]) \subseteq \mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])[p]$ is of finite index, $\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])[p]$ is also finite. By the same reasoning, the conclusion follows. □

Proposition 3.2. *Under (Im), if* $\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])$ *is* Λ*-cotorsion, then* $\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty])$ *has no proper* Λ*-submodule of finite index. Thus, we have*

$$\mathrm{char}_\Lambda \mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty]) = \mathrm{Fitt}_\Lambda \mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty]),$$
$$\mathrm{Sel}_{\mathrm{min}}(K_\infty, E[p^\infty]) \simeq \mathrm{Sel}_{N^-}(K_\infty, E[p^\infty]).$$

Proof. This follows from [Gre99, Prop. 4.14], which covers the cyclotomic case actually, but the argument generalizes to our setting as mentioned in the proof of [PW11, Prop. 3.6]. □

The following corollary follows from (3.1) and the above two propositions.

Corollary 3.3. *Under (Im) and (Ram), if* $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])$ *is* Λ*-cotorsion with vanishing of* μ*-invariant, we have isomorphisms*

$$\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty]) \simeq \mathrm{Sel}_{\min}(K_\infty, E[p^\infty]) \simeq \mathrm{Sel}_{\mathrm{Gr}}(K_\infty, E[p^\infty]).$$

4. The proof of the main theorem via Iwasawa theory

We first gather some tools from Iwasawa theory and give a proof of Theorem 1.1.

4.1. **Iwasawa theory.** The anticyclotomic main conjecture for (E, p, K) is now completely known for our setting.

Theorem 4.1. *Under (Im),(Ram),(Spl),(Na), and (Def), we have the following statements:*

(1) $L_p(E/K_\infty)$ *is non-zero.*
(2) $\mu(L_p(E/K_\infty)) = 0$.
(3) $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])$ *is* Λ*-cotorsion with vanishing of* μ*-invariants.*
(4) $(L_p(E/K_\infty)) = \mathrm{char}_\Lambda\left(\mathrm{Sel}(K_\infty, E[p^\infty])^\vee\right)$.

Proof. (1) It is proved in [Vat02].
(2) It is proved in [Vat03].
(3) This follows from (1), (2), and the Euler system divisibility

$$(L_p(E/K_\infty)) \subseteq \mathrm{char}_\Lambda\left(\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])^\vee\right)$$

obtained from the bipartite Euler system argument [BD05, PW11]. Condition (Na) is implicitly used in the Euler system argument. See [KPW17, Assu. 1.1 and Rem. 1.4] for this issue.
(4) By using (3) and Corollary 3.3, we can identify $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])$ with the minimal Selmer group. By [Gre99, Prop. 2.1], the minimal Selmer group also coincides with the classical Selmer group. The opposite divisibility

$$(L_p(E/K_\infty)) \supseteq \mathrm{char}_\Lambda\left(\mathrm{Sel}(K_\infty, E[p^\infty])^\vee\right)$$

follows from [SU14]. Condition (Spl) is needed only for this last statement to invoke [SU14].

□

Corollary 4.2. *Under (Im),(Ram),(Na), and (Def), the classical Selmer group* $\mathrm{Sel}(K_\infty, E[p^\infty])$ *has no proper* Λ*-submodule of finite index; thus,*

$$\mathrm{char}_\Lambda \mathrm{Sel}(K_\infty, E[p^\infty]) = \mathrm{Fitt}_\Lambda \mathrm{Sel}(K_\infty, E[p^\infty]).$$

Proof. By Theorem 4.1.(3), $\mathrm{Sel}_{N^-}(K_\infty, E[p^\infty])$ is Λ-cotorsion. By Proposition 3.1, $\mathrm{Sel}_{\min}(K_\infty, E[p^\infty])$ is also Λ-cotorsion. The conclusion follows from Proposition 3.2 and the identification of the minimal Selmer group and the classical Selmer group. □

We recall the control theorem.

Proposition 4.3 (Control theorem)**.** *Let* $\omega_n = \omega_n(X) = (1+X)^{p^n} - 1 \in \mathbb{Z}_p[\![X]\!] \simeq \Lambda$*. Under (Im), (Na), and (Def), the restriction map*

$$\mathrm{Sel}(K_n, E[p^\infty]) \to \mathrm{Sel}(K_\infty, E[p^\infty])[\omega_n]$$

is injective with the finite cokernel whose size is bounded independently of n*. If we further assume (Ram), then it is an isomorphism.*

Proof. See [CH15, Prop. 1.9] with the identifications of Selmer groups in Corollaries 3.3 and 4.2. □

4.2. **The proof of Theorem 1.1.** By the anticyclotomic main conjecture (Theorem 4.1), we have

$$(L_p(E/K_\infty)) = \mathrm{char}_\Lambda\left(\mathrm{Sel}(K_\infty, E[p^\infty])^\vee\right).$$

By Corollary 4.2, the above equality becomes

$$(L_p(E/K_\infty)) = \mathrm{Fitt}_\Lambda\left(\mathrm{Sel}(K_\infty, E[p^\infty])^\vee\right)$$

Under the quotient map $\Lambda \to \Lambda_n = \Lambda/\omega_n$, it becomes

$$((\theta(f_\alpha/K_n)\cdot\iota(\theta(f_\alpha/K_n)))) = \mathrm{Fitt}_{\Lambda_n}\left((\mathrm{Sel}(K_\infty, E[p^\infty])[\omega_n])^\vee\right)$$

since Fitting ideals are compatible with base change. By using the functional equation of theta elements (2.4) and the control theorem (Proposition 4.3), we have

$$(\theta(f_\alpha/K_n))^2 = \mathrm{Fitt}_{\Lambda_n}\left(\mathrm{Sel}(K_n, E[p^\infty])^\vee\right).$$

Theorem 1.1 now follows from Lemma 2.2, and the ideal is principal thanks to the above equality.

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